Sharp Signed-Profile Costs and Finite Error Certificates for Fourfold Ergodic Averages

Dr. Muralidhara Subbarao
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Stony Brook University
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Abstract

We study the October 4 manuscript on pointwise convergence of fourfold ergodic averages, with particular attention to its four changed triples and its use of arbitrary-system lower-length convergence. For the signed smooth profiles with three vanishing moments required by its appendix, we prove a sharp leading support–accuracy law: approximation of $c^{-1}\mathbf{1}_{(0,c]}$ by a profile supported in $[-R, R]$ costs $3c/(2R) + O(R^{-2})$ in L^1, uniformly for $1 \le c \le 2$. A four-node signed correction, subsequently smoothed, matches a cubic Chebyshev dual lower bound. We derive an explicit finite-scale smooth-to-Cesaro oscillation error, explain why approximate moments do not provide the source's compact primitive, and quantify finite-test near-saturation and imperfect axis orthogonality. These results do not prove the imported local analytic estimates, all-order mixing, or global pointwise convergence. Classical moment duality, transference, and projection identities are credited, and the previous M292 packing improvement is not counted again. Contribution assessment: C3/5, a notable source-specific sharp profile cost and quantitative transfer certificate, with limited finite-test activation controls and no global theorem certification.

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1 Source statement and dependency boundaries

The source [1] claims that for every invertible, bimeasurable, mixing probability-preserving transformation T and every fixed bounded complex tuple (f_1, f_2, f_3, f_4),

$$\frac{1}{N}\sum_{n=1}^{N}\prod_{j=1}^{4} f_j(T^{jn}x) \longrightarrow \prod_{j=1}^{4}\int f_j\,d\mu \quad \text{a.e.,} \qquad N \to \infty \text{ through all integers.} \tag{1}$$

The null set can depend on the tuple. No mixing rate or standardness of the original probability space is assumed. The source first codes the fixed tuple and its integer translates into a standard factor. Our complete literal reading covers all ten cached public texts, including all seven scientific sections; it is not a full proof audit of the external inputs.

The auxiliary extensions need not be mixing or ergodic. Their proper-slot laws require pointwise convergence for *all* triples

$$\mathcal{J} = \{\{1,2,3\},\{1,2,4\},\{1,3,4\},\{2,3,4\}\} \tag{2}$$

on arbitrary invertible probability-preserving systems. Neither a theorem only for $(n, 2n, 3n)$ nor a norm-limit theorem supplies this interface.

Source section	Mechanism and retained limitation
01 Introduction	Fixed-function, all-length fourfold assertion; companions explicitly named.
02 Structure	Imported all-order mixing; nonergodic distal structure, relative weak mixing and disjointness; proved relative norm replacement.
03 Selected laws	Failure-dependent lengths in a suspension; actual selected laws, functoriality, all proper marginals given the full distal array.
04 Packing	Gaussian tensor-packing proof with a sparse exceptional graph. Its optimization belongs to prior M292, not the present contribution.
05 Channels	Three output groups, pair marginals preserved after selection, and a weighted conditional density for the full triple.
06 Activation	Exclude three output roles by iterated axes and Bessel; saturate five datum types; regroup outputs; prove square nonvanishing.
07 Lower length	Changed-coefficient checks for four role sets; imported H/P local proofs; local-to-orbit oscillation; signed profiles and all-length passage.

Here H denotes the source's companion *An L^3 bound for the trilinear Hilbert transform*; P denotes its companion *Pointwise convergence of triple ergodic averages for mixing transformations*. We have not independently audited H's Sections 3–10 or P's depth-varying analytic proof. The source does more than simply cite Hilbert-transform boundedness: its appendix checks altered charts, lifts, counting, and depth scheduling. That mechanism must be preserved, but cannot be independently certified by the finite coefficient identities below.

2 Changed slopes and exact cancellation

For $J \in \mathcal{J}$, let $S = \{0\} \cup J$. The four-linear local form is

$$|K|^{-2}\int_{\mathbb{R}^2} w_K(x,t)\prod_{j\in S} z_j(x+jt)\,dx\,dt.$$

The source requires interior support in K for each role coordinate, Gevrey-type derivative bounds after scaling by $|K|$, and the exact line cancellations $\int w_K(v-jt,t)\,dt = 0$ for every $j \in S$. The following algebra explains some, but not all, of the slope adaptation.

Proposition 2.1 (Finite role identities, credited to interpolation)**.** *In increasing role order the primitive vectors are*

S	(c_j)				$\operatorname{diam} S$
0123	-1	3	-3	1	3
0124	-3	8	-6	1	4
0134	-1	2	-2	1	4
0234	-1	6	-8	3	4

They satisfy $\sum_j c_j j^a = 0$ *for* $a = 0, 1, 2$*. If* $\sum_j m_j = \sum_j j m_j = 0$*, then* $m_j = c_j(v + jv')$ *for some real* v, v'*, and* $\sum_j m_j^2/c_j = 0$*. The offset region* $|\xi + j\eta| < e$ *for all* $j \in S$ *has area* $4e^2/\operatorname{diam} S$*.*

Proof. The coefficient of degree three in the Lagrange interpolation formula for a polynomial of degree at most two is zero. Thus c_j is proportional to $\prod_{l \neq j}(j - l)^{-1}$; clearing denominators and dividing their common factor gives the table. The vectors c and (jc_j) are independent, belong to the two-dimensional nullspace of the two displayed constraints, and therefore span it. Substitution proves the quadratic identity. For the area, the intermediate inequalities follow from the extreme ones by convexity. The map $(\xi, \eta) \mapsto (\xi + s_-\eta, \xi + s_+\eta)$ has determinant $s_+ - s_-$ and maps the region onto $(-e, e)^2$. □

The source's primitive denominator list is cleared by 24 in its quadratic randomization. Its height base changes from 6 to 12, and its annular multiplier from 36 to 144: for nonzero differences α_i between the roles, $|\alpha_i| \mid 12$, so $144/\alpha_i \in 12\mathbb{Z}$ and $12\alpha_j/\alpha_i \in \mathbb{Z}$. These calculations do not establish the source's Weyl alternative, inverse theorem, progression-counting estimate, or depth-varying Bessel estimate. We credit the table to the source and classical interpolation.

2.1 Why all three moments must vanish exactly

An averaging profile in the source is a real $C_c^\infty(\mathbb{R})$ function with mass one, moments of orders one through three equal to zero, and $\|\varphi^{(n)}\|_\infty \le (G(n+2))^{G(n+2)}$ for a profile-dependent $G \ge 1$. Write $\varphi_A(t) = A^{-1}\varphi(t/A)$.

Proposition 2.2 (Compact primitive and a residual obstruction)**.** *Let* $\varphi \in C_c^\infty(\mathbb{R})$ *have mass one and moments* r_1, r_2, r_3*. For* $b = \varphi_2 - \varphi$*, its canonical fourth primitive is*

$$g(t) = \frac{1}{6}\int_{-\infty}^{t} (t - u)^3 b(u)\, du.$$

It is compactly supported if and only if $r_1 = r_2 = r_3 = 0$*. To the right of the support of* b *its value is exactly*

$$g(t) = -\tfrac{1}{2}r_1 t^2 + \tfrac{3}{2}r_2 t - \tfrac{7}{6}r_3. \tag{3}$$

Proof. The moments of b are $0, (2^a - 1)r_a$ in orders $a = 1, 2, 3$. Expand $(t - u)^3$ to obtain (3). The primitive vanishes to the left of the support; to vanish also on a right half-line its three polynomial coefficients must vanish. Differentiating four times gives b. □

In the source's local-to-orbit transfer, with u a unit-integral bump, one takes $F(z, \sigma) = u(z)D^{-3}g(D\sigma)$ and

$$\mathcal{W} = \prod_{j \in S} (\partial_\sigma - j\partial_z) F.$$

Along the line $z = v - j\sigma$, one factor is the line derivative, giving the required cancellation by compact support. Integrating in z removes every z derivative and gives $Db(D\sigma)$. For $r = DA$ this is precisely

$$\frac{1}{r}\int \mathcal{W}(z, t/r)\, dz = \varphi_{2A}(t) - \varphi_A(t).$$

This is the correct normalization for every role set. A tiny moment error does not give a tiny compact primitive: it leaves the polynomial tail in (3). The remaining local-to-orbit estimate is still conditional on the source's changed-coefficient depth theorem.

3 A sharp support cost for signed profiles

Let $h_c(t) = c^{-1}\mathbf{1}_{(0,c]}(t)$, $1 \le c \le 2$. For $R \ge c$ define $E_c(R)$ as the infimum of $\|\varphi - h_c\|_1$ over the source's averaging profiles supported in $[-R, R]$. The constant G may depend on the profile; no uniform Gevrey constant is imposed. All integrals in this section are Lebesgue integrals or integrals of explicit finite signed measures. The total variation norm of a signed measure is its full mass $|\nu|(\mathbb{R})$, not half that mass.

Theorem 3.1 (Dual lower bound)**.** *Every such profile satisfies*

$$\|\varphi - h_c\|_1 \ge L_c(R) := \frac{3c}{2R} - \frac{c^3}{R^3}. \tag{4}$$

If instead its first and third moments are r_1, r_3, the right side can be replaced by $L_c(R) - 3|r_1|/R - 4|r_3|/R^3$.

Proof. The cubic Chebyshev polynomial $T_3(x) = 4x^3 - 3x$ has modulus at most one on $[-1, 1]$, as follows from $T_3(\cos\theta) = \cos 3\theta$. The moment conditions and $\int t h_c = c/2$, $\int t^3 h_c = c^3/4$ imply

$$\int T_3(t/R)(\varphi - h_c)(t)\, dt = L_c(R).$$

Both functions are supported in $[-R, R]$, so the absolute integral is at most their L^1 distance. With imperfect moments the additional term is $4r_3/R^3 - 3r_1/R$; the triangle inequality gives the stated bound. □

This is a classical moment-dual witness. Its source-specific force is the factor three in the leading support cost, rather than the weaker $c/(2R)$ obtained by testing only the first moment.

Lemma 3.2 (Four-node exact signed repair)**.** *Fix $c \in [1, 2]$, $A \ge 8c$, and $0 \le v \le A^{-4}$. Set $m_1 = c/2$, $m_2 = c^2/3 + v$, $m_3 = c^3/4$, and define*

$$u_1 = -\frac{4m_2}{3A^2}, \qquad u_{1/2} = \frac{4m_2}{3A^2},$$
$$v_1 = \frac{m_1}{3A} - \frac{4m_3}{3A^3}, \qquad v_{1/2} = -\frac{8m_1}{3A} + \frac{8m_3}{3A^3}.$$

At nodes $\pm A, \pm A/2$ assign weights $w_{\pm A} = (u_1 \pm v_1)/2$ and $w_{\pm A/2} = (u_{1/2} \pm v_{1/2})/2$; let ν be the resulting signed atomic measure. Then

$$\int d\nu = 0, \qquad \int t^j\, d\nu = -m_j \quad (j = 1, 2, 3), \qquad \|\nu\|_{\mathrm{TV}} = L_c(A). \tag{5}$$

Among all signed measures on $[-A, A]$ with these four moments, ν has minimum total variation.

Proof. Adding and subtracting the weights reduces the moment equations to $u_1 + u_{1/2} = 0$, $A^2(u_1 + u_{1/2}/4) = -m_2$, $A(v_1 + v_{1/2}/2) = -m_1$, and $A^3(v_1 + v_{1/2}/8) = -m_3$; direct substitution proves them. The odd differences dominate the even sums. Indeed

$$3A^3(v_1 - |u_1|) = m_1A^2 - 4m_2A - 4m_3.$$

We have $m_2 \le c^2/3+1/4096 < c^2/2$. Writing $A = ct$, $t \ge 8$, the right side is at least $c^3(t^2/2-2t-1) > 0$. Similarly,

$$3A^3(-v_{1/2} - |u_{1/2}|) = 8m_1A^2 - 4m_2A - 8m_3 \ge c^3(4t^2 - 2t - 2) > 0.$$

The weights at $-A, -A/2, A/2, A$ consequently have signs $-, +, -, +$, respectively. Their absolute sum is $v_1 - v_{1/2} = L_c(A)$. These are exactly the signs of $T_3(t/A)$ at those nodes. Any signed measure with the same moments satisfies $\int T_3(t/A)\, d\nu = 3m_1/A - 4m_3/A^3 = L_c(A)$, so the total variation dual bound proves optimality, independently of its second moment. □

Theorem 3.3 (Smooth repair and sharp leading cost)**.** *Fix an even, nonnegative, unit-integral Gevrey-order-two bump $\psi \in C_c^\infty([-1,1])$. Put $b_\psi = \int |t|\psi(t)\, dt \le 1$, $v_\psi = \int t^2\psi(t)\, dt \le 1$, and $V_\psi = \|\psi'\|_1$. For $A \ge 8c$ and $0 < \sigma \le A^{-2}$ use Lemma 3.2 with $v = \sigma^2 v_\psi$, and set*

$$\varphi = (h_c(t)\, dt + \nu) * \psi_\sigma, \qquad \psi_\sigma(t) = \sigma^{-1}\psi(t/\sigma). \tag{6}$$

This is an averaging profile supported in $[-A-\sigma, A+\sigma]$, and

$$\|\varphi - h_c\|_1 \le L_c(A) + 2\sigma b_\psi/c, \tag{7}$$

$$\mathrm{Var}(\varphi) = \|\varphi'\|_1 \le 2/c + L_c(A)V_\psi/\sigma. \tag{8}$$

Uniformly for $1 \le c \le 2$, as $R \to \infty$,

$$E_c(R) = \frac{3c}{2R} + O_\psi(R^{-2}), \qquad \lim_{R\to\infty} RE_c(R) = \frac{3c}{2}. \tag{9}$$

Proof. Convolution of finite signed measures obeys the binomial moment formula. The measure $h_c\, dt + \nu$ has mass one, first and third moments zero, and second moment $-\sigma^2 v_\psi$. The even smoothing measure has first and third moments zero and second moment $\sigma^2 v_\psi$. Their convolution therefore has exactly the required moments, including the third: its cross term is three times the first moment of the first measure times the second moment of the second, hence zero. The support assertion follows by addition of supports.

Translations of h_c satisfy $\|h_c(\cdot - u) - h_c\|_1 = 2\min(|u|/c, 1)$. Integration against ψ_σ proves the smoothing error bound. Convolution by a probability measure is a total variation contraction, so the correction costs at most $\|\nu\|_{\mathrm{TV}} = L_c(A)$. Moreover, $(h_c * \psi_\sigma)' = (\psi_\sigma(t) - \psi_\sigma(t-c))/c$ and $\|\nu * \psi_\sigma'\|_1 \le L_c(A)V_\psi/\sigma$, proving (8).

For completeness the usual bump proportional to $\exp[-1/(1-t^2)]$ on $|t| < 1$, zero outside, has $\|\psi^{(n)}\|_\infty \le C^{n+1}(n!)^2$. Indeed at distance δ from an endpoint a complex disk of radius $c_0\delta$ has real part of $1/(1-z^2)$ at least c_1/δ; Cauchy's estimate bounds the derivative by $n!(C/\delta)^n e^{-c_1/\delta}$. Maximizing in δ gives the stated bound, with the interior absorbed in C; the same estimates give smooth zero extension at the endpoints. Scaling this bump and convolving with our finite measure yields the source's derivative bound for some finite $G = G(A, c, \sigma, \psi)$. It is not uniform in A.

For $R \ge 17$ choose $\sigma = R^{-2}$ and $A = R - \sigma$. Then $A \ge 8c$ and $\sigma \le A^{-2}$. The upper bound is $L_c(R - R^{-2}) + 2R^{-2}/c = 3c/(2R) + O(R^{-2})$ uniformly in c. Theorem 3.1 supplies the lower bound with error $O(R^{-3})$. □

3.1 An exact example and its interpretation

For $c = 1$, $A = 8$, $v = 0$, the weights at $(-8, -4, 4, 8)$ are

$$\frac{1}{9216}(-125, 794, -730, 61).$$

Their moments are $(0, -1/2, -1/3, -1/4)$ and their variation is $95/512 = 3/16 - 1/512$. This atomic measure is an exact algebraic example, not a smooth averaging profile. Smoothing it without changing its second moment would leave a nonzero moment $\sigma^2 v_\psi$. Formula (6) adjusts the even weights first and avoids that error. Nonnegative profiles cannot do the job: a nonnegative mass-one measure with second moment zero is concentrated at zero, so it has no smooth density.

The source already obtains $O(1/A)$ error by a qualitative four-bump moment correction. The new increment here is the optimal leading constant and the simultaneous support, moment, and variation certificate, not the existence of signed corrections. We do not assert a new general moment-duality theorem or attainment of $E_c(R)$ by a smooth extremizer. The exact finite-radius atomic optimization is a different problem from the smooth infimum.

4 Finite-scale changed-slope error bounds

Let R be any invertible probability-preserving transformation and let $\|g_j\|_\infty \le 1$. For $J \in \mathcal{J}$ write

$$a_n^J(x) = \prod_{j \in J} g_j(R^{jn}x), \quad B_k^J(x) = \frac{1}{\lfloor c2^k \rfloor} \sum_{n=1}^{\lfloor c2^k \rfloor} a_n^J(x), \quad V_k^{\varphi,J}(x) = \sum_{n \in \mathbb{Z}} 2^{-k}\varphi(2^{-k}n) a_n^J(x).$$

The latter sum is finite. Negative indices cause no difficulty because R is invertible. All statements below are uniform in the system and in these bounded functions; there is no mixing assumption.

Proposition 4.1 (Sampling certificate)**.** *For $k \ge 0$, a compactly supported smooth profile, and $\varepsilon = \|\varphi - h_c\|_1$, put $B_\varphi = 3/c + \mathrm{Var}(\varphi)$. Then pointwise*

$$|B_k^J - V_k^{\varphi,J}| \le \varepsilon + B_\varphi 2^{-k}. \tag{10}$$

For the profiles above, $B_\varphi \le 5/c + L_c(A)V_\psi/\sigma$.

Proof. Put $H = 2^{-k}$, $N = \lfloor c/H \rfloor$. The sum with weights $Hh_c(nH)$ has precisely the indices $1, \dots, N$, and its difference from the normalized Cesaro sum is at most $1 - NH/c \le H/c$. For a compactly supported function F of bounded variation,

$$\left| H \sum_{n \in \mathbb{Z}} F(nH) - \int F \right| \le H \,\mathrm{Var}(F).$$

To see this, compare $F(nH)$ with its values in each cell $((n-1)H, nH]$, integrate the variation on that cell, and sum. Use the endpoint convention $h_c(0) = 0$ and $h_c(c) = 1/c$ specified above; the same estimate includes both jumps. With $F = |h_c - \varphi|$, we have $\mathrm{Var}(F) \le 2/c + \mathrm{Var}(\varphi)$. Multiplication by $|a_n^J| \le 1$ and the triangle inequality now prove (10). □

Corollary 4.2 (Conditional block-oscillation certificate)**.** *Assume, for this fixed profile, the source's imported-input-dependent estimate*

$$\sum_{b=0}^{W-1} \int \max_{k_b \le l \le k_{b+1}} |V_l^{\varphi,J} - V_{k_b}^{\varphi,J}| \, d\mu \le C_\varphi \sqrt{W} \tag{11}$$

for every increasing deterministic integer list $k_0, \dots, k_W$. If $k_0 \ge K \ge 0$, then

$$\sum_{b=0}^{W-1} \int \max_{k_b \le l \le k_{b+1}} |B_l^J - B_{k_b}^J| \, d\mu \le C_\varphi \sqrt{W} + 2W(\varepsilon + B_\varphi 2^{-K}). \tag{12}$$

In particular the expected fraction of blocks with this maximum greater than $t > 0$ is at most $C_\varphi/(t\sqrt{W}) + 2(\varepsilon + B_\varphi 2^{-K})/t$. For the union over all four J, add the four bounds.

Proof. Each difference changes by at most twice the right side of (10), uniformly over its block. Integrate, sum, and apply (11); Markov's inequality gives the last claim. □

With $\sigma = A^{-2}$ the two explicit terms are $\varepsilon = O(A^{-1})$ and $B_\varphi = O_\psi(A)$. Thus $2^K \ge A^2$ makes their sum $O_\psi(A^{-1})$. This is a useful finite-scale comparison, not a rate for (1): C_φ and the source's local analytic constants are not controlled by this calculation. Fixing a profile, taking the oscillation limit, and then improving profile accuracy is legitimate; silently varying the profile with the block count is not.

4.1 Bounded input approximation on all four triples

This auxiliary classical estimate records precisely the topology needed to extend a bounded dense test class. It is not supplied by norm convergence alone. If g_j, g_j' are bounded by one, put $e_j = \|g_j - g_j'\|_2$. For $t > 0$,

$$\mu\left\{\sup_{N \ge 1} |A_N^J(g) - A_N^J(g')| > t\right\} \le \frac{(\sum_{j \in J} e_j^{2/3})^3}{t^2} \le \frac{9 \sum_{j \in J} e_j^2}{t^2}, \tag{13}$$

where A_N^J denotes the unsmoothed average. Indeed telescoping the product and Cauchy–Schwarz on each orbit give the upper bound $\sum_j (M_{R^j} |g_j - g_j'|^2)^{1/2}$, where M_U is the positive ergodic maximal operator for U. Its weak $(1,1)$ inequality, valid for every probability-preserving U, bounds the exceptional probability by $\sum e_j^2/t_j^2$ for $\sum t_j = t$. Minimization gives $t_j = t e_j^{2/3} / \sum e_i^{2/3}$, with zero-error terms omitted; Holder's inequality gives the last bound. A union over the four J is legitimate. No assertion that a characteristic-factor subspace is dense in the whole L^2 space is involved. This estimate is consistent with the earlier Ram comparison [4]; we do not count it as new.

5 Finite activation certificates

The source's activation is not a mere choice of a joining with product marginals. It compares the actual selected laws through functorial extensions. On its base system let P denote the full distal parameter array. Its five datum types are

$$\mathcal{D}_0 = \sigma(P), \qquad \mathcal{D}_l = \sigma(P, y_i : i \ne l), \quad 1 \le l \le 4.$$

Pullback makes each datum type increasing along an extension tower. The following local statement is valid on any common probability extension, independent of the source's existence machinery.

Proposition 5.1 (Near-saturation with a finite test bank)**.** *For nested sigma fields $\mathcal{A} \subset \mathcal{B}$ and $f \in L^2$,*

$$\|\mathbb{E}(f \mid \mathcal{B}) - \mathbb{E}(f \mid \mathcal{A})\|_2^2 = \|\mathbb{E}(f \mid \mathcal{B})\|_2^2 - \|\mathbb{E}(f \mid \mathcal{A})\|_2^2. \tag{14}$$

If the current energy is within ϵ of the supremum over all admissible further extensions for this test and datum type, every such extension changes its conditional expectation by at most $\sqrt{\epsilon}$ in L^2. Pairing that change with any $h \in L^2$ costs at most $\|h\|_2 \sqrt{\epsilon}$.

For a finite bank $(f_i, l_i)_{i=1}^M$, along a process in which each chosen extension increases at least one bank energy by more than $\epsilon > 0$, there can be at most $\lfloor \sum_i \|f_i\|_2^2/\epsilon \rfloor$ such extensions. At a terminal stage with no possible gain greater than ϵ, the preceding bounds hold simultaneously for this finite bank. If $\|f - f_i\|_2 \le \delta$ for the same datum type, the bound becomes $\sqrt{\epsilon} + \delta$ for f.

Proof. Conditional expectations are orthogonal projections. For nested projections $P \le Q$, the difference $Q - P$ is an orthogonal projection orthogonal to P. Pythagoras proves (14), and Cauchy–Schwarz proves the pairing estimate. The total potential $\sum_i \|P_{l_i} f_i\|_2^2$ is monotone along every extension and bounded by $\sum_i \|f_i\|_2^2$. Every genuine update increases it by more than ϵ, proving the update bound. If updates were always possible beyond that bound, choosing one at each stage would contradict the potential bound. Thus the finite existential procedure has a terminal stage. Finally $\|Q - P\|_{2\to 2} \le 1$, which proves the approximation assertion with one δ, not two. □

The supremum is over the declared admissible extension class, not all abstract sigma fields; pullback preserves all norms used above. This argument gives no computable oracle for finding a profitable extension or recognizing termination. A finite bank is not uniform saturation on the unit ball of L^2. The source uses a countable dense bank and an inverse-limit tower to obtain exact identities for further extensions; that infinite construction is not replaced by this finite certificate. In particular L^2 errors for these tests are not total variation errors between the whole conditional laws.

Proposition 5.2 (Imperfect axis-orthogonality test)**.** *In a real or complex Hilbert space suppose $\|H\| \le B$, $\|H_a\| \le K$ for $1 \le a \le n$, $|\langle H_a, H_b\rangle| \le \eta$ for $a \ne b$, and $|\langle H, H_a\rangle - c| \le \delta$. Then*

$$n(|c| - \delta)_+^2 \le B^2\{K^2 + (n-1)\eta\}. \tag{15}$$

Consequently, if $0 \le \eta \le K^2$ and $(|c| - \delta)_+^2 > B^2\eta$, then

$$n \le \frac{B^2(K^2 - \eta)}{(|c| - \delta)_+^2 - B^2\eta}.$$

Proof. The Gram matrix of the H_a has operator norm at most $K^2 + (n-1)\eta$: its quadratic form is bounded by $K^2 \sum |z_a|^2 + \eta \sum_{a\ne b} |z_a||z_b|$, and $(\sum |z_a|)^2 \le n \sum |z_a|^2$. The analysis operator therefore gives $\sum_a |\langle H, H_a\rangle|^2 \le B^2\{K^2 + (n-1)\eta\}$. The correlation hypotheses lower-bound this sum, proving both assertions. □

This is a robust form of the exact Bessel contradiction in source Section 06, where iterated axes have the same nonzero correlation and are orthogonal. It is useful only when B, K, δ, η are verified for the actual axes. It supplies neither their existence nor errors uniform in their number. The source's functorial line laws and proper slot independence justify those axes in its exact argument; an abstract finite Gram computation alone does not do so.

5.1 What the square still needs

The subsequent square argument needs its two marginal laws separately. The column law is obtained by functoriality; the row law uses the single-state marginal identity. It is not an assumed row–column symmetry. Saturation of $\mathcal{D}_0$ fixes the old column law after extra distal parameters are adjoined. Saturation of the appropriate missing-state $\mathcal{D}_l$, together with proper three-row laws, gives row locality. Deterministic parameter reanchoring then gives the conditional product law needed for the Hilbert-space nonvanishing test. These exact interfaces must all be retained when applying Proposition 5.1; one cannot substitute small errors for them without propagating errors in each of these other steps.

For example, three uniform signs conditioned to have product $+1$ have independent pairs but are not jointly independent. Likewise four uniform signs conditioned to have product $+1$ have all proper product marginals but not a product joint law. These classical examples explain why the selected-channel density and the source's activation closure cannot be replaced by a proper-marginal argument. We do not attribute that shortcut to the source, which provides additional mechanisms.

6 Original Ram evidence and current literature

The fresh original readings used here concern finite moment methods, not a private proof of fourfold convergence. Volume 21's extremal chapter [2] proves convex reduction and separating-polynomial bounds for positive measures, and propagates finite moment errors through dual coefficients. Its transforms chapter distinguishes finite Taylor information from an unjustified infinite transform or joint-law claim. The signed correction here is not a positive representing measure; its total variation dual argument has been proved explicitly above. Thus positivity-based moment existence is not silently transferred to it.

The original TR248 passages [3] use the entire moment Gram matrix, not only its zeroth entry, and retain Taylor remainders when assembling finite matrices. Their pertinent lesson here is to repair the smoothing variance explicitly and to distinguish exact moments from small residuals. TR168 [4] is a later comparative report, not foundational evidence for the new arbitrary-system changed triples. The prior M292 report's packing improvement is a separate result: it can be transferred only after matching its tensor and channel premises, and receives no new contribution credit in M293.

As of the October 8, 2026 public-primary check, Kuca's March 2026 survey [5] concerns joint *norm* ergodicity. The September 2026 version of Kosz–Mirek–Peluse–Wan–Wright [6] addresses distinct-degree polynomial pointwise averages; its stated hypothesis does not include four consecutive linear slopes. These abstract-level checks do not establish or refute the source's fixed-slope pointwise claim. Jamneshan's version-four primary record [7] explicitly reports incorporation of a corrigendum concerning the relative structural theory. The source already accounts for that correction, including the closure distinction for relative modules; no new structural gap is asserted here. We have not audited the full external structural proofs. Classical transference is associated with Calderon [8]; the quantitative sampling and block error bounds above were proved directly, without treating bibliographic metadata as a proof.

7 Advantages, costs, and verification scope

The sharp profile construction exposes a real constraint on the source's approximation: an error at most ε needs support of leading size at least $3c/(2\varepsilon)$ in this symmetric support model. The four-node correction attains that leading cost while preserving exact moments after smoothing. The variation estimate then exposes the finite sampling cost, which a qualitative existence argument leaves hidden. The primitive calculation detects a particularly dangerous shortcut: small moment defects create a noncompact polynomial tail, not the compact annihilating kernel used in the analytic input.

The costs are equally explicit. The profiles are signed; their derivative constants grow as their smoothing scale shrinks. Our bound on B_φ does not bound the deep analytic constant C_φ. Finite-test saturation is existential, and finitely many conditional moments do not determine a full selected law. Axis error tolerances must remain small relative to the correlation, rather than merely tend to zero for each separately fixed axis. No finite-dimensional computation certifies the arbitrary-depth analytic estimates, all-order mixing, or the final infinite activation argument.

Useful future questions are a joint optimization of support, variation and derivative constants; comparable exact moment repairs for other role sets; and a complete error propagation through the actual square interfaces when quantitative conditional-law data are independently available. More vanishing moments suggest other Chebyshev dual witnesses, but this report proves only the four-role, three-moment result stated above. It makes no historical-priority claim for the classical tools.

7.1 Reading and executable checks

The author literally read all 2,804 lines of the ten cached public texts, including all 2,562 lines of the seven scientific sections, in bounded untruncated displays. Fresh private-original reading covers 189 lines of Volume 21's extremal chapter, 165 lines of its transforms chapter, and TR248 lines 1030–1121 (92 lines): 446 original lines in total. TR168 was read at lines 44–117, 216–272 and 347–470 (255 lines), with its unread complements recorded. Other corpus files and the H/P deep companion proofs have not been read in full for this report. Before/after hashes, actual intervals, literature scope, tests and final page images are recorded in the owned M293 evidence files.

The accompanying exact rational checks exercise role cancellations, signed-repair moments and signs, smoothing moments, the primitive tail, sampling endpoints, finite projection energies and finite Gram bounds. They supplement the proofs above, not the global source theorem. The author-run exact suite passed 7,739 parameter cases and 32,045 assertions using rational and integer arithmetic. These comprise 484 role cases, 1,125 signed repairs, one worked atomic example, 585 smoothing moment cases, 625 primitive-tail cases, 363 endpoint cases, 54 bounded variation quadrature cases, 2,500 conditional projection cases, 2,000 axis Gram cases and two proper-marginal examples. None is counted as the parent's independently run tests. The bounded variation fixtures are not claimed to be the smooth Gevrey profiles of the theorem.

References

[1] *Pointwise convergence of fourfold ergodic averages for mixing transformations*, October 4, 2026. Cached source at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`; public source directory.

[2] M. Subbarao, Ram Calculus Books, Volume 21, Chapter 02, original *extremal* and *transforms* proof files. Confidential local corpus; exact paths and read intervals in the local ledger.

[3] M. Subbarao, *Stable Adaptive Ram Frames*, TR248, original local manuscript, lines 1030–1121.

[4] M. Subbarao, TR-item-168-astra, later local comparison of triple ergodic averages; scoped reading only, as listed above.

[5] B. Kuca, *Joint ergodicity - 40 years on*, arXiv:2603.18974, March 19, 2026. Abstract and version metadata checked, not a full proof audit.

[6] D. Kosz, M. Mirek, S. Peluse, R. Wan and J. Wright, *The multilinear circle method and a question of Bergelson*, arXiv:2411.09478v4, September 2, 2026. Primary abstract/version record; see its precise scope.

[7] A. Jamneshan, *An uncountable Furstenberg–Zimmer structure theory*, arXiv:2103.17167v4, September 27, 2025, incorporating the stated corrigendum.

[8] A. P. Calderon, *Ergodic theory and translation-invariant operators*, Proc. Natl. Acad. Sci. USA 59 (1968), 349–353, primary journal record.